Paper Title: Forecasting X with Y — A Comparative Evaluation

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Abstract

Liquidity management ensures that a bank has sufficient cash to fulfill its short-term obligations while minimizing the cost of holding excess liquidity, and non-maturity deposits (NMDs) are among the most important sources of this liquidity. Despite their economic importance, forecasting NMD volumes remains methodologically underdeveloped, with no established consensus on the optimal modeling approach. This study compares naive benchmarks, a regularized linear model, gradient-boosted trees, a feedforward neural network, and a Temporal Fusion Transformer for forecasting German household overnight deposit volumes over a 100-day horizon, using a rolling-origin evaluation across four distinct interest-rate regimes between 2009 and 2025. The regularized linear model achieved the lowest mean forecast error (MAE = 9.36 Bn. €) and was the only model with a positive mean R^2 (0.178), outperforming the naïve random-walk benchmark (MAE = 11.78 Bn. €) by 20.5%, while more complex non-linear architectures did not consistently improve on this simple benchmark. These results indicate that increased model complexity does not necessarily translate into higher forecasting accuracy for this task, with direct implications for how banks might select forecasting tools for liquidity management.

Contents

1 Introduction

The Basel Committee on Banking Supervision (BCBS, 2016) defines non-maturity deposits (NMDs) as liabilities without a contractual maturity where the depositor is free to withdraw at any time. They consist mainly of fully transferable deposits and non-transferable deposits that can be converted by end of the next business day. The interest paid from NMDs is significantly below other products issued by the banks. Moreover, since there is no fixed maturity date, banks reserve the right to change this interest rate at any time. Therefore, because of their stability and low-cost, NMDs are an important source of funding for banks. Especially for German banks, NMDs nowadays represent the most important funding source after exhibiting a dynamic growth in the aftermath of the financial crisis of 2008 (Schlueter et al., 2015). Considering this information, it becomes clear how important is the study of NMDs deposits for banks liquidity management, the process of assessing the liquidity needs of a bank and supplying or absorbing the appropriate amount of liquidity through open market operations (European Central Bank, 2011). However, because of their stochastic nature, the study of these products become, to say the least, extremely challenging. Despite its importance, the academic research on this topic is still scarce. Recent years have seen a growing interest on the subject in the form of graduate theses, journals and papers (Winkler, 2025; Marena et al., 2022). Nevertheless, this current literature presents a wide range of approaches, from economic to statistical and mathematical models, there is yet to be a general accepted scientific consensus to the optimal methodology for modeling NMDs. Hence, the need for a structured and effective methodology for the forecasting of these financial products is evident.

This study addresses the identified research gap by developing a comparative framework for forecasting non-maturity deposit volumes. We evaluate naive benchmarks, statistical models, and machine-learning models across alternative covariate sets and multiple forecast origins representing different economic environments. Although the study is motivated by NMD modelling, the empirical analysis uses publicly available German household overnight deposit data as a proxy for aggregate household NMD volumes.

The study makes three contributions. First, it examines whether greater model complexity leads to more accurate forecasts. Second, it evaluates how interest-rate, macroeconomic, and financial-market covariates affect forecasting performance. Third, it investigates whether model performance remains stable across different economic periods. All models are assessed using a common rolling-origin design and a realistic information set in which realized economic covariates beyond the forecast origin are unavailable.

The remainder of this paper is structured as follows. Section 2 provides related literature on NMD modeling and forecasting, and positions this study within this research. Section 3 presents the data and preprocessing methodology, including variable definitions, interpolation, and target transformation. Section 4 describes the forecasting framework, the models under comparison, and the rolling-origin evaluation design. Section 5 reports and compares the forecasting results. Section 6 discusses these findings and their managerial implications. Finally, Section 7 concludes the paper and outlines directions for future research.

Research Question(s):

RQ1 How accurately can statistical and machine-learning models forecast German household overnight deposit volumes over a 100-day horizon?

RQ2 Do nonlinear machine-learning models outperform linear and naive benchmark models under the same rolling-origin evaluation framework?

RQ3 Which groups of covariates provide the most improvement in forecasting performance?

2 Related Work

Table 1: Positioning of this paper relative to related NMD research.

Reference	Modeling Approach	Rolling-Origin Evaluation
Winkler (2025)	Lévy-driven OU process vs. LSTM	✗
This paper	Naive, Ridge, XGBoost, FFN, TFT comparison	✓

3 Data and Preprocessing

3.1 Target Variable and Explanatory Variables

Detailed non-maturity deposits (NMDs) data are generally not publicly available because they contain sensitive bank information. For this study, we use publicly available end-of-period volume of overnight deposits held by private households, published by the Deutsche Bundesbank (Deutsche Bundesbank, 2025a) as a proxy for aggregate NMD volumes. Household overnight deposits provide an accurate publicly observable proxy since they are a subset of NMDs and account for a substantial share of the banks' total deposit funding (Deutsche Bundesbank, 2024). Therefore, the aggregate volume of household overnight deposits is likely to respond to several of the same economic drivers. The dataset covers the period January 2003 to September 2025 and the series is measured in billions of euros.

Euro Short-Term Rate (€STR): sourced from the ECB (European Central Bank, 2025b), is available at daily frequency and captures the average overnight borrowing rate of banks in the euro area. The reasoning for including €STR as a covariate is that a shift from it could cause a shift in the rates banks apply to deposit accounts, meaning a change in the total overnight deposits.

German geopolitical risk index (GPRC_DEU): published on a monthly basis (Caldara and Iacoviello, 2022), quantifies the geopolitical risk in Germany by using automated text search in the U.S. newspapers *New York Times*, *Chicago Tribune* and *Washington Post*. Geopolotical context may have an influence because as uncertainty rises, customers might be more drawn to postpone spending and leaving cash in their accounts, leading to an increase in overnight deposits.

Month-on-month inflation (MoM-Inflation): sourced from Destatis (Destatis, 2025), calculates the percentage change in the Consumer Price Index relative to the preceding month. Inflation changes how much households spend and save, therefore it could also lead to explain movements of their total overnight deposits.

DAX daily closing prices: obtained via the yfinance Python library (Aroussi, 2025), retrieving data from Yahoo Finance (Yahoo Finance, 2026) for the ticker `^GDAXI`. As DAX reflects the changes in the German stock market as well as investor confidence, this may influence the total amount hold by private households.

10-year German government bond yield (10Y Bond): sourced at business-day frequency from the Deutsche Bundesbank (Deutsche Bundesbank, 2025b). As the 10-year old government bond is a relatively safe investment product, private household may be drawn to invest more on them if the rate is sufficiently attractive, investing less in overnight deposits account.

Deposit Rate: refers to the household deposit interest rate, sourced from the Deutsche Bundesbank (Deutsche Bundesbank, 2025a). The more interest households receive on deposits, the more attractive it may be to keep their money in overnight deposit accounts rather than seeking alternative investments.

3.2 Data Alignment and Transformation: Interpolation Methods

Since multiple variables are published at monthly frequency, they therefore require interpolation to a daily grid before model training. Deposit volume, the deposit rate, inflation, and geopolitical risk, are then converted from their monthly series to daily frequency following piecewise cubic Hermite interpolating polynomial (PCHIP) interpolation Winkler (2025, Ch. 5.2). This preserves local monotonicity of the input series and avoids artificial overshoots between monthly observations (Fritsch and Carlson, 1980).

As each published value reflects the previous business day's transactions, it constitutes a direct, transaction-based measure of interbank market conditions. While €STR was officially introduced in October 2019, the sample period extends back to January 2003. For the period March 2017 to September 2019, the Pre-€STR (European Central Bank, 2025a) is used as a direct predecessor series. For earlier observations, the series is reconstructed following Winkler (2025, Ch. 5.2).

For variables available at business-day frequency, (€STR, DAX and 10Y Bond), weekend and public holiday gaps are handled via forward-fill, carrying the last observed value forward to the following business day. This ensures a complete daily index across the full sample period while avoiding the introduction of artificial variation on non-trading days.

The following figure illustrates illustrates all variables over the full sample period after interpolation to daily frequency. As evident from Figure 1, the deposit-volume level exhibits a clear upward trend throughout the timeline, indicating nonstationarity in the mean. To this first differencing is applied, to obtain a stationary target series and to limit accumulating directional errors (Box et al., 2015), as illustrated in Figure 2. Second differencing was rejected do to over-differencing.

After interpolation to daily frequency, the dataset comprises $8,276$ observations covering the full sample period January 2003 to September 2025.

4 Forecasting Models

4.1 Forecasting Framework

Let y_t denote the volume of German household overnight deposits at time t, and let $\mathbf{x}_t$ denote the vector of covariates. The objective is to forecast deposit volumes over a 100-day horizon using the information available during the preceding 30 days.

At time of forecast t, the forecasting models use a lookback window of $L = 30$. The information set therefore contains the historical deposit volume and the selected covariates over the period $t - L + 1, \ldots, t$. Each model produces the complete vector of predicted changes for the following $H = 100$ days:

$$\widehat{\boldsymbol{\Delta y}}_{t+1:t+H} = f(y_{t-L+1:t}, \mathbf{x}_{t-L+1:t}), \qquad L = 30, \quad H = 100. \tag{1}$$

The study therefore applies a direct multi-output forecasting strategy. Each model estimates all 100 future changes simultaneously rather than recursively one day at a time. The predicted changes are then converted back into deposit-volume levels according to

$$\widehat{y}_{t+h} = y_t + \sum_{j=1}^{h} \widehat{\Delta y}_{t+j}, \qquad h = 1, \ldots, H.$$

Information available up to time t are used for model estimation and feature scaling. Actual covariate values for those 100 days are not used. For TFT model, while covariates are held constant to the most recent observed value, date-based variables known in advance are provided for the forecast horizon.

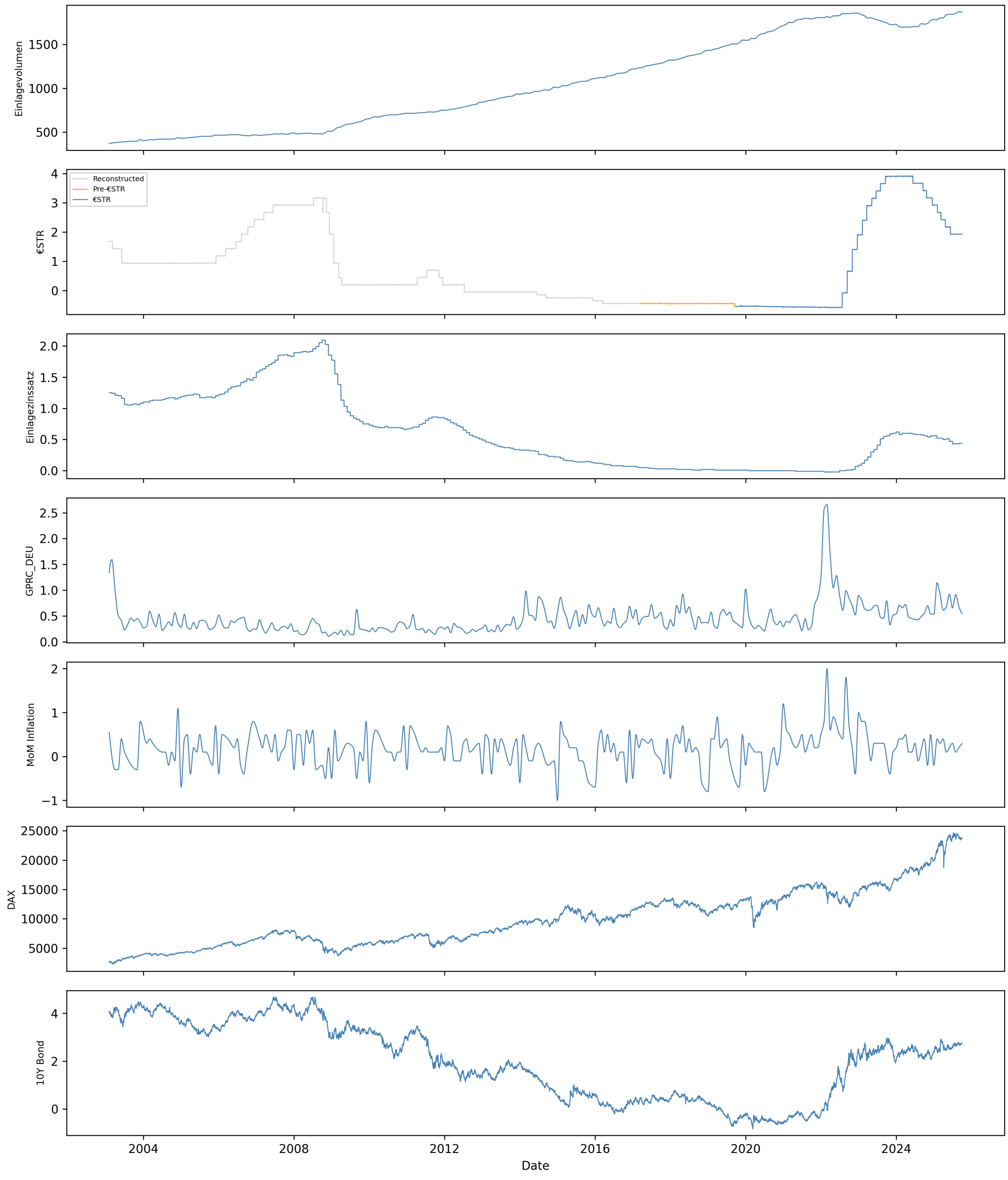

Figure 1: Time series of all variables over the sample period January 2003 – December 2025. All monthly series are shown after PCHIP interpolation to daily frequency. The €STR series begins in October 2019 following its introduction by the ECB.

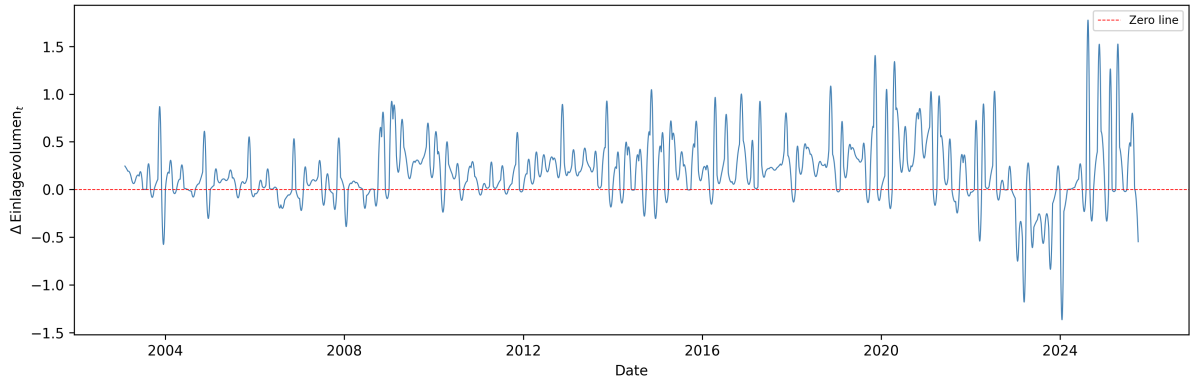

Figure 2: First difference of Einlagevolumen$_t$ over the full sample period January 2003 – December 2025. The series fluctuates around zero throughout, confirming stationarity after differencing. Increased volatility from 2022 onward coincides with the ECB interest rate tightening cycle.

4.2 Forecasting Models

Naive benchmarks. Random Walk extrapolates the last observed daily change at a constant rate,

$$\hat{y}_{t_0+h} = y_{t_0} + h \cdot \Delta y_{t_0}, \qquad h = 1, \ldots, 100,$$

which corresponds to an ARIMA$(0,1,0)$ process with drift. Naive Hold instead assumes the level remains constant, $\hat{y}_{t_0+h} = y_{t_0}$. Both benchmarks are fully interpretable and require no covariates, serving as a reference against to which compare the more complex models.

Linear Regression (Ridge). A multi-output ridge regression maps the flattened 30-day lookback window to the 100-step forecast vector, regularized by an ℓ_2 penalty α selected via time-series cross-validation. As a linear model, Ridge provides interpretable results.

XGBoost. Gradient-boosted regression trees are trained on the same flattened input representation, with hyperparameters (max depth, learning rate, subsample ratio) (Chen and Guestrin, 2016). These hyperparameters are selected using chronological time-series cross-validation when tuning is enabled. Boosted trees can capture non-linear covariate interactions that Ridge cannot (Hastie et al., 2009, Ch. 10), while remaining moderately interpretable.

Feedforward Neural Network (FFN). The untuned FFN with two hidden layers (128, 64 units, ReLU activation) is trained to directly predict the 100-step output vector. Training uses Adam optimizer with early stopping based on the last 15% of the training window, chronologically. FFN trades interpretability for greater representational flexibility.

Temporal Fusion Transformer (TFT). Unlike the sequence models above, the TFT explicitly distinguishes between *known* future inputs (calendar features: weekday, month-end indicator, day-of-month) and *unknown* future inputs (the six macro-financial covariates). The model is trained with early stopping (patience 5 validation checks) on validation loss. Despite being the most complex model considered, the TFT's attention mechanism was explicitly designed by Lim et al. (2021) to offer a degree of interpretability that purely black-box models typically lack.

Each sequence model is fitted S times with different random seeds to form an ensemble. When $S > 1$, repeated model fits are combined by averaging their forecasts. Linear regression, FFN, and XGBoost use bootstrap-resampled training samples and different random seeds, while repeated TFT fits use different random seeds..

4.3 Covariate Specifications

To assess the predictive value of different economic information, each non-naive model is evaluated using the nine predefined covariate sets shown in Table 2. These specifications compare economically related variable groups, individual interest-rate variables, and parts of the complete dataset.

Table 2: Covariate specifications

Specification	Included variables
All	€STR, deposit rate, geopolitical risk, inflation, DAX, and ten-year German government bond yield
Interest rates	€STR and deposit rate
Financial markets	DAX and ten-year government bond yield
Macroeconomic	Geopolitical risk and inflation
Rates and macroeconomy	€STR, deposit rate, geopolitical risk, and inflation
Excluding DAX	All variables except DAX
Excluding inflation	All variables except inflation
€STR only	€STR
Deposit rate only	Deposit rate

4.4 Rolling-Origin Evaluation

Forecasting performance is assessed using an expanding-window rolling-origin design. At each forecast origin o_k, all observations available up to that date form the training sample, while the following $H = 100$ days form the evaluation window:

$$\mathcal{T}_k = \{1, \dots, o_k\}, \qquad \mathcal{E}_k = \{o_k + 1, \dots, o_k + H\}.$$

The models are re-estimated at every origin as newly observed data becomes available. The evaluation uses four origins: 1 January 2009, 1 January 2013, 1 March 2019, and 1 June 2025. The models are re-estimated at every origin as newly observed data becomes available. The evaluation uses four origins: 1 January 2009, 1 January 2013, 1 March 2019, and 1 June 2025. These dates were selected to represent distinct interest-rate and macroeconomic regimes – the 2008/2009 financial crisis, the subsequent low-interest-rate period, the pre-COVID environment, and the post-2022 high-interest-rate tightening cycle – rather than a dense monthly rolling scheme, in order to explicitly test model robustness across qualitatively different economic conditions.

4.5 Evaluation

Performance is evaluated for both the predicted first differences and the reconstructed deposit-volume levels. Let $y_{k,h}$ and $\widehat{y}_{k,h}$ denote the observed and predicted volume at horizon h after origin o_k. The evaluation metrics are

$$\mathrm{MAE}_k = \frac{1}{H} \sum_{h=1}^{H} \left| y_{k,h} - \widehat{y}_{k,h} \right|,$$

$$\mathrm{RMSE}_k = \sqrt{\frac{1}{H} \sum_{h=1}^{H} \left(y_{k,h} - \widehat{y}_{k,h} \right)^2},$$

$$R_k^2 = 1 - \frac{\sum_{h=1}^{H} \left(y_{k,h} - \widehat{y}_{k,h} \right)^2}{\sum_{h=1}^{H} \left(y_{k,h} - \overline{y}_k \right)^2},$$

where $\overline{y}_k$ is the mean observed volume in evaluation window k. Negative R^2 values indicate performance below the test-window mean benchmark.

Metrics are calculated separately at each forecast origin and summarized by their mean and standard deviation across origins. The primary comparison uses MAE for the reconstructed deposit-volume levels, measured in billions of euros. RMSE and R^2 provide complementary information, while metrics on first differences are used as diagnostic measures. A Diebold–Mariano test for pairwise significance of forecast accuracy differences was implemented but not applied to the main results, as the four rolling origins used here provide insufficient statistical power for a reliable significance test; model comparisons are therefore reported descriptively rather than as formal hypothesis tests.

5 Results

5.1 Forecasting Performance

Table 3 reports the best-performing covariate specification for each model according to mean level MAE. The results are descriptive because the covariate sets are selected using the same four evaluation windows on which performance is reported.

Table 3: Forecasting performance across four rolling origins. MAE and RMSE are measured in EUR billions and reported as mean ± standard deviation.

Model	Best covariate set	MAE	RMSE	Mean R^2
Linear model	All covariates	9.36 ± 11.11	10.26 ± 12.08	0.178
TFT	Macroeconomic	10.78 ± 12.61	12.71 ± 14.18	−0.361
Last-change extrapolation	None	11.78 ± 12.81	14.41 ± 14.25	−0.564
FFN	Macroeconomic	11.85 ± 6.90	15.22 ± 10.35	−0.605
XGBoost	Excluding inflation	12.71 ± 13.28	14.90 ± 14.79	−0.784
Constant-level benchmark	None	17.38 ± 10.13	21.03 ± 10.52	−2.246

The linear model achieved the lowest average MAE and was the only approach with a positive mean R^2. TFT ranked second, followed by the last-change benchmark. The FFN and XGBoost did not improve upon this simple benchmark on average, while the constant-level forecast produced the largest errors. Therefore, greater model complexity did not consistently lead to higher forecasting accuracy.

5.2 Effect of Covariate Selection

The preferred covariate specification differed across models. The linear model performed best with all six explanatory variables, whereas the FFN and TFT obtained their lowest MAE using only geopolitical risk and inflation. XGBoost performed best when inflation was excluded. These findings suggest that the predictive value of a covariate depends on how the model represents linear and nonlinear relationships. They should not, however, be interpreted as evidence of causal effects.

5.3 Variation Across Forecast Origins

Forecasting performance varied substantially across the four evaluation periods, as reflected by the large standard deviations in Table 3. The FFN displayed the lowest variation in MAE, but its average error remained above those of the linear model and TFT. The results indicate that no model dominated at every forecast origin and that model rankings depend on the economic environment.

Because the evaluation contains only four forecast origins, the results should be interpreted as a comparison across selected historical periods rather than as evidence of statistical superiority. Nevertheless, the benchmark shows that simple and interpretable methods remain competitive for forecasting aggregate household overnight deposits.

6 Discussion

The empirical findings from our rolling-origin benchmark provide crucial insights into how distinct algorithmic architectures handle highly autocorrelated, noisy, and regime-shifting financial time series over a 100-day forecasting horizon. Contrary to common expectations in modern machine learning literature, our evaluation across four distinct macroeconomic interest rate regimes (2009–2025) reveals that **simpler, regularized linear models outperform complex non-linear architectures**.

This section critically evaluates the performance, structural strengths, and architectural constraints of all six benchmarked models based on the empirical results presented in Table 3.

6.1 Ridge Regression: The Gold Standard for Stability

Ridge achieved the highest overall accuracy (MAE = 9.36 Bn. €), outperforming Random Walk by 20.5% and remaining the **only model to achieve a positive mean R^2 (0.178)**.

The structural advantage of L_2-regularized linear models in cumulative horizon forecasting lies in their smooth penalty mechanism. When predicting daily volume differences (Δy_t) over $H = 100$ steps, small daily estimation errors compound linearly or quadratically during final volume reconstruction:

$$\widehat{y}_{t_0+h} = y_{t_0} + \sum_{j=1}^{h} \widehat{\Delta y}_{t_0+j} \tag{2}$$

The Ridge penalty shrinks non-essential feature weights toward zero, effectively filtering out high-frequency interbank noise. Consequently, when supplied with the full feature space (`All covariates`), Linear Regression maintained superior stability across major interest rate pivots.

6.2 Temporal Fusion Transformer (TFT): Architectural Overhead vs. Input Constraints

The Temporal Fusion Transformer ranked second overall (MAE = 10.78 Bn. €). Although TFT is designed specifically for multi-horizon time series forecasting via self-attention mechanisms, its full performance potential was constrained by the strict ex-post evaluation setup.

To prevent look-ahead bias, future values of dynamic covariates were held constant using a naive-hold assumption over the 100-day horizon. As a result, TFT's temporal self-attention layers received static input sequences for dynamic economic drivers, neutralizing its primary structural advantage for capturing future macro trajectories. While TFT successfully outperformed tree ensembles and the naive benchmark, its parameter overhead provided no benefit over Ridge under static future covariate assumptions.

6.3 Feedforward Neural Networks (FFN) and XGBoost: Overfitting and Extrapolation Traps

Neither FFN (MAE = 11.85 Bn. €) nor XGBoost (MAE = 12.71 Bn. €) managed to consistently outperform the simple Random Walk benchmark (MAE = 11.78 Bn. €) on average:

- **FFN Consistency vs. Precision:** Interestingly, the FFN exhibited the lowest standard deviation across forecast origins (±6.90 Bn. €), indicating high cross-regime consistency.

However, its average MAE remained slightly worse than the simple benchmark, showing that unregularized dense layers tend to fit noise in daily volume differences.

- **XGBoost Extrapolation Limitation:** Decision trees partition the feature space into orthogonal hyper-rectangles and cannot extrapolate continuous trends beyond the bounded range of historical training data. When deposit volumes reached new record levels (such as during the 2022–2025 rate tightening cycle), XGBoost capped predictions at historical leaf thresholds, causing systematic underestimation.

6.4 Understanding the Out-of-Sample R^2 Metric in Time Series

As observed in Table 3, several models yielded negative R^2 values. In multi-step time series forecasting, R^2 evaluates predictions against the localized mean of the test evaluation window ($\overline{y}_k$). Because $\overline{y}_k$ is calculated *post-hoc* using future realizations unavailable at the forecast origin, it represents an unrealistically strong benchmark. A negative R^2 indicates that a model's multi-step trajectory diverged from this post-hoc mean, rather than indicating zero predictive utility. Therefore, Level MAE serves as our primary metric for practical model ranking.

6.5 Managerial Implications and Practical Takeaways

For bank Asset-Liability Management (ALM) and treasury operations, predictability, computational speed, and interpretability are critical. Our benchmark demonstrates that complex deep learning models do not automatically translate into superior deposit forecasts when future macro-covariates are uncertain. **Ridge provides the most robust, interpretable, and accurate framework for 100-day NMD volume forecasting.**

7 Conclusion and Outlook

Accurate forecasting of non-maturity deposit (NMD) volumes is fundamental for bank liquidity management, regulatory balance sheet planning, and interest rate risk assessment. However, modeling aggregate NMDs is challenged by non-stationarity, high autocorrelation, and macroeconomic regime shifts.

This study evaluated six forecasting approaches across four distinct interest rate regimes (2009–2025) using an expanding-window rolling-origin setup. Our core conclusions address our three research questions:

1. **Model Complexity vs. Accuracy (RQ1 & RQ2):** Greater model complexity does not guarantee higher forecast accuracy. Ridge achieved the best overall performance (MAE = 9.36 Bn. €, $R^2 = 0.178$), outperforming FFN (MAE 11.85 Bn. €) and XGBoost (MAE 12.71 Bn. €).

2. **Covariate Selection (RQ3):** The predictive value of covariates depends heavily on model architecture. Linear models benefit from broad, multi-variable feature sets (`All covariates`) where regularized weights filter noise, whereas non-linear models performed best when restricted to core macroeconomic indicators (Inflation and Geopolitical Risk).

3. **Practical ALM Utility:** For bank liquidity buffer management (e.g., LCR requirements), simple regularized models offer minimal computational overhead, complete transparency, and robust protection against regime shifts.

Outlook and Future Work. Future research could extend this comparative framework by incorporating scenario-based path projections for dynamic covariates rather than holding them constant over the forecast horizon. Additionally, combining linear baseline trends with non-linear residual models (hybrid architectures) may better capture sudden systemic liquidity shocks while preserving long-term trend stability.

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A Appendix: Additional Figures and Tables

B Appendix: Code

Listing 1: Example: model fitting excerpt

```
def fit_sequence_model(model_name, X_train, y_train, seed=SEED):
    # ...
    pass
```